

MD MOMINUL HAQUE

mdmominulhq@outlook.com | Google Scholar | LinkedIn: mominulhq | Dhaka, Bangladesh

Profile

AI researcher with research interests in Computer Vision, Deep Learning, Medical Image Analysis, and Image Processing, with a focus on explainable (XAI) and trustworthy AI, multimodal learning, and test-time adaptation (TTA). Experienced in developing robust and resource-efficient deep learning models using PyTorch and TensorFlow, with an emphasis on generalization, interpretability, and real-world imaging applications.

Education

Universiti Teknologi PETRONAS (UTP) (QS 2026: 251 | THE: 201-250) Perak, Malaysia
Master of Science in Information Technology (By Research) 2024 – June 2026

Funding: Fully funded under the UTP Graduate Assistantship (GA) Scheme

Thesis: *MVT-Net: A Lightweight and Explainable Hybrid CNN-Transformer Framework for Retinal Vessel Segmentation*

Stamford University Bangladesh (Top 5% of Class) Dhaka, Bangladesh
Bachelor of Science in Computer Science & Engineering CGPA: 3.80 / 4.00

Thesis: *Attention-based Long-term Recurrent Convolutional Network for Physiotherapy Exercise Classification*

Technical Skills & Research Competencies

Programming: Python, C, C++

DL/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn

Libraries: OpenCV, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Albumentations

Web & Databases: HTML, CSS, SQL

Languages: English (IELTS 6.5), Bengali

DL & CV Methodologies: Image & Semantic Segmentation, Object Detection, Explainable AI (XAI / Grad-CAM++), Transfer Learning, Test-Time Adaptation (TTA), Attention Mechanisms

Tools & Systems: Git/GitHub, Overleaf (LaTeX), VS Code, Jupyter, Linux, Windows, Cloud Platforms

Experience

Graduate Teaching Assistant Sep 2024 – June 2026
Universiti Teknologi PETRONAS Perak, Malaysia

AI Engineer Jul 2024 – Sep 2024
Eravend Innovation Private Ltd. Dhaka, Bangladesh

Research Assistant (Remote) Mar 2024 – Sep 2024
Canada Syntex Machine Learning Alberta, Canada

Computer Science Teacher Dec 2023 – Mar 2024
World School of Bangladesh Gazipur, Bangladesh

Artificial Intelligence (AI) Intern Aug 2023 – Nov 2023
Inflexionpoint Technologies BD Ltd. Dhaka, Bangladesh

Junior Software & Systems Engineer July 2022 – July 2023
Biosense Ltd. Dhaka, Bangladesh

Leadership & Awards

Executive Committee – International Affairs and Student Mobility 2025 – Present
Student Representative Council (SRC), Universiti Teknologi PETRONAS (UTP)
Elected student leader as ambassador for international students in UTP.

PETRONAS Focused Recognition Award May 2026
Centre for Excellence in Teaching and Learning (EdEx), Universiti Teknologi PETRONAS
Awarded for technical leadership in designing and piloting the UTP Staff Continuous Professional Development (CPD) Dashboard, recognized for driving core institutional cultural beliefs and impacting academic career progression tracking.

Best Presenter Awards
– 2025 IEEE 9th International Conference on Software Engineering & Computer Science (ICSECS 2025), Malaysia.
– 2024 IEEE 9th International Conference on Computational Intelligence and Applications (ICCIA 2024), China.

Manuscript Reviewer – Elsevier Biomedical Signal Processing and Control (Q1 Journal)
Reviewed scientific manuscripts related to deep learning and biomedical image analysis.

Publications

- [1] Md Mominul Haque, Saipunidzam Mahamad, Suziah Sulaiman, Abdullateef Oluwabemiga Balogun, and Hussaini Mamman. “ToxoSegFusion: Attention-enhanced Dual-backbone Neural Architecture for Retinal Lesion Segmentation”. In: *IIUM Engineering Journal* 27.1 (2026). (**SJR Quartile: Q3**), pp. 321–347. DOI: 10.31436/iiumej.v27i1.3806.
- [2] Md Mominul Haque, Saipunidzam Mahamad, Suziah Sulaiman, and Abdullateef Oluwabemiga Balogun. “Noise-Space Adaptive Multi-Domain Filtering: A Unified Framework for Low-Light Urban Surveillance and Medical Retinal Imaging”. In: *Applied Research* 5.3 (2026). (**SJR Quartile: Q2**), e70106. DOI: <https://doi.org/10.1002/app1.70106>.
- [3] Md Mominul Haque, Saipunidzam Mahamad, Suziah Sulaiman, Abdullateef Oluwabemiga Balogun, and Kanneh Momoh. “DAFNet-Res: A Hybrid-Attentive Decoder Network with CLAHE-Enhanced Supervision for Retinal Vessel Segmentation”. In: *2025 IEEE 9th International Conference on Software Engineering Computer Systems (ICSECS)*. 2025, pp. 326–331. DOI: 10.1109/ICSECS65227.2025.11279243.
- [4] Saipunidzam Mahamad, Yi Han Chin, Nur Izzah Nasuha Zulmuksah, Md Mominul Haque, Muhammad Shaheen, and Kanwal Nisar. “Technical Review: Architecting an AI-Driven Decision Support System for Enhanced Online Learning and Assessment”. In: *Future Internet* 17.9 (2025). (**SJR Quartile: Q1**). ISSN: 1999-5903. DOI: 10.3390/fi17090383.
- [5] Md. Mominul Haque, Sweetly Akter, and Adnan Ferdous Ashrafi. “SwinMedNet: Leveraging Swin Transformer for Robust Diabetic Retinopathy Classification from the RetinaMNIST2D Dataset”. In: *2024 6th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)*. 2024, pp. 1286–1291. DOI: 10.1109/ICEEICT62016.2024.10534544.
- [6] Md Mominul Haque, Mohammed Al-Mukhtar, MD Samiul Islam, Alhassan Alie Kamara, and Mehedi Hasan Nirob. “A Case Study of Human Segmentation in Multispectral Imaging”. In: *2024 IEEE 9th International Conference on Computational Intelligence and Applications (ICCIA)*. 2024, pp. 155–160. DOI: 10.1109/ICCIA62557.2024.10719107.
- [7] Manik Hossain, Sumaiya Binte Siraj, Md. Mominul Haque, Gihun Song, Byungyong Ryu, and Md. Tauhid Bin Iqbal. “Physiotherapy Exercise Classification by Exploring Knowledge Distillation with Attention Modules”. In: *2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*. 2023, pp. 1–11. DOI: 10.1109/NCIM59001.2023.10212747.

Manuscripts Under Review

- **Md Mominul Haque**, Adnan Ferdous Ashrafi, Saipunidzam Mahamad, & Sweety Akter. “Swin-MedNetV2: A Hierarchical Transformer for Cross-Modal Medical Image Classification,” *Neural Computing and Applications*. (SJR Quartile: Q1)
- **Md Mominul Haque**, Hosna Asma Ull, & Nur Alam MD. “Null-Space Gradient Projection for Bounded Test-Time Adaptation in Echocardiographic Segmentation,” *Biomedical Signal Processing and Control*. (SJR Quartile: Q1)

Projects

CPD Learning Hours Dashboard for UTP Staff - A live dashboard developed for Universiti Teknologi PETRONAS (UTP). The system provides real-time analytics and reporting on training hours, compliance tracking, and professional development progress.

Face Mask Detection – Developed real-time face mask detection.

Face Recognition – Developed real-time facial recognition for EraVend GmbH.

Real-Time Face Mask Detection – Implemented MobileNetV2 for real-time mask detection.

Object Detection (Custom Data, YOLO API) – Built domain-specific YOLO object detector.

Facial Expression Recognition – Trained CNN model using PyTorch and custom datasets.

Bangla Handwritten Digit Recognition – Implemented Inception-GoogLeNet on Ekush dataset.

Online Crime Reporting System – Web-based platform to improve public crime reporting.

Departmental Shop Management – Inventory and sales management system.

Licenses & Certifications

Graduate Technologist (Information & Communication Technology) – Malaysia Board of Technologists (MBOT)

Aug 2026

Registration No: GT26080165

IBM AI Engineering Professional Certificate (In Progress) – Coursera

Deep Learning Specialization – Coursera

DeepLearning.AI TensorFlow Developer – Coursera

Generative Adversarial Networks (GANs) Specialization – Coursera

Neural Networks and Deep Learning – Coursera

Computer Vision Basics – Coursera

Cloud Computing Basics (Cloud 101) – Coursera